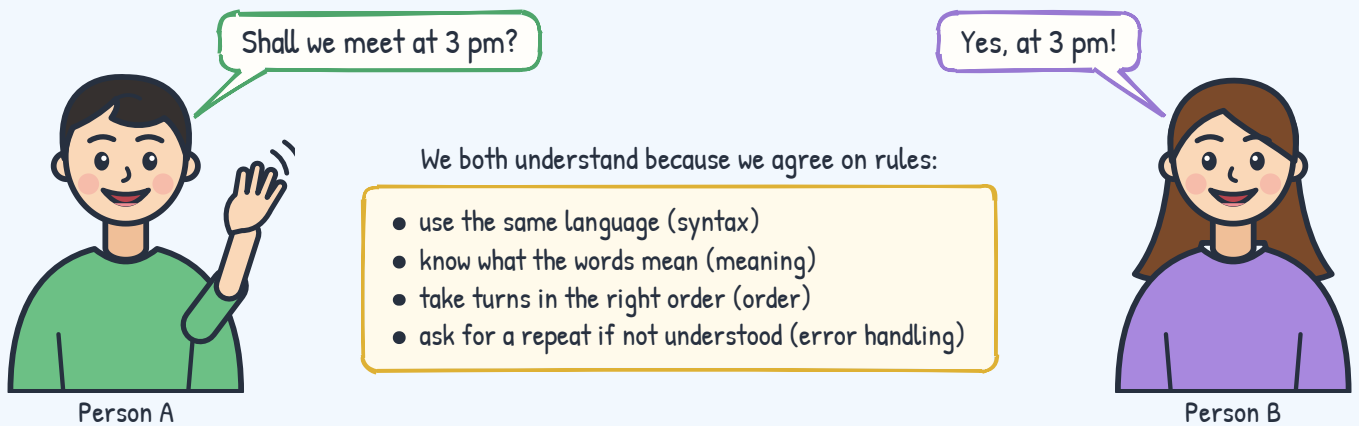


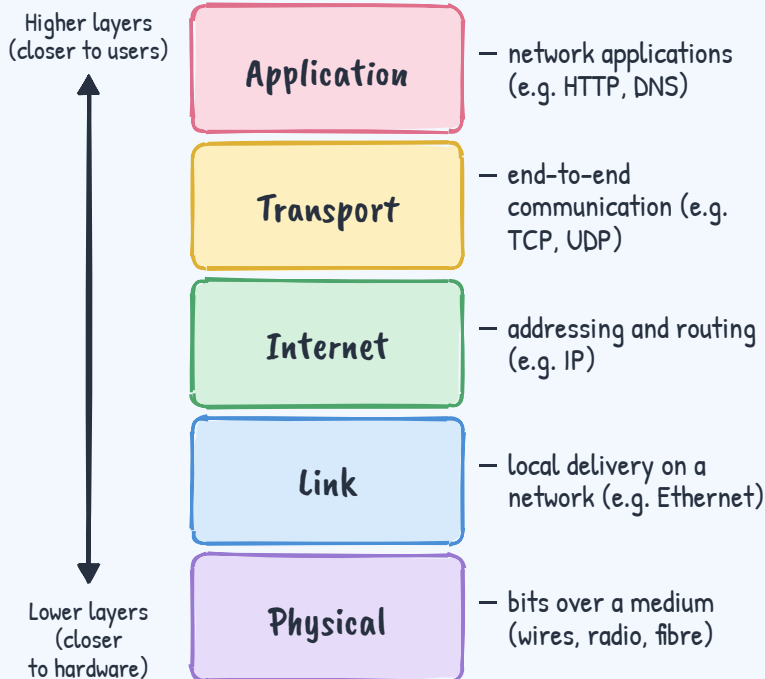
10. PROTOCOLS: AGREED RULES

A CONVERSATION ANALOGY



Protocols let independent devices interpret the same exchange.

A LAYERED PROTOCOL STACK



KEY IDEAS



SYNTAX

Format of the data. E.g. message fields, headers, character encoding.



MEANING

What the data represents. E.g. `GET` means "request a web page".

1 2 3

ORDER

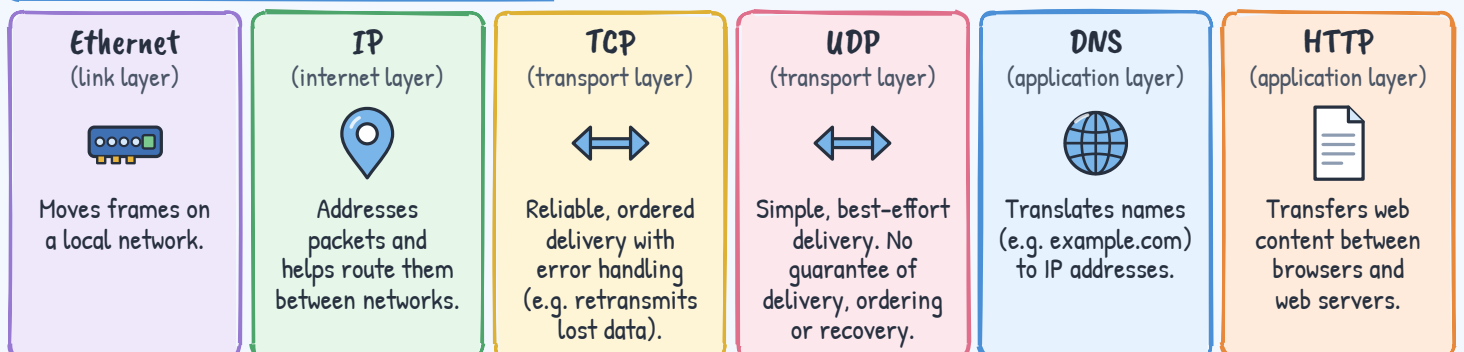
Steps must happen in a specific sequence. E.g. connect before sending data.



ERROR HANDLING

Detect problems and recover. E.g. checksums, timeouts, retransmission, error messages.

EXAMPLES OF COMMON PROTOCOLS



REMEMBER: Protocols are agreements. Without them, devices could not understand each other.